

From Discount Retailer to Retail Giant: The Target Case Study

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

Answer:

The data type of all the columns in the “customers” table are as follows.

Column Name	Data Type
customer_id	STRING
customer_unique_id	STRING
customer_zip_code_prefix	INTEGER
customer_city	STRING
customer_state	STRING

The data types can be seen by clicking on the “columns” table in the database and then clicking the SCHEMA.

The screenshot displays a data exploration tool interface. On the left, the 'Explorer' panel shows a tree view of workspace resources. Under 'scaler-target-project-dsml', the 'Target_Business_Case_Study' folder is expanded, revealing several tables including 'customers', 'geolocation', 'order_items', 'order_reviews', 'orders', 'payments', 'products', and 'sellers'. The 'customers' table is selected and highlighted. The main panel on the right shows the 'customers' table selected in the top bar. Below this, there are tabs for 'SCHEMA', 'DETAILS', 'PREVIEW', and 'LINEAGE'. The 'SCHEMA' tab is active, displaying a table with columns: 'Field name', 'Type', 'Mode', 'Key', 'Collation', and 'Default value'. The rows list the columns of the 'customers' table: 'customer_id' (STRING, NULLABLE), 'customer_unique_id' (STRING, NULLABLE), 'customer_zip_code_prefix' (INTEGER, NULLABLE), 'customer_city' (STRING, NULLABLE), and 'customer_state' (STRING, NULLABLE). At the bottom of the schema view, there are two buttons: 'EDIT SCHEMA' and 'VIEW ROW ACCESS POLICIES'.

Field name	Type	Mode	Key	Collation	Default value
customer_id	STRING	NULLABLE			
customer_unique_id	STRING	NULLABLE			
customer_zip_code_prefix	INTEGER	NULLABLE			
customer_city	STRING	NULLABLE			
customer_state	STRING	NULLABLE			

2. Get the time range between which the orders were placed.

Answer:

The time range between which the orders were placed is as follows:

- The first order was placed on 4th September 2016
- The last order was placed on 17th October 2018

The Query is as follows:

```
select min(order_purchase_timestamp) as First_Order,  
       max(order_purchase_timestamp) as Last_Order  
from `Target_Business_Case_Study.orders`
```

The screenshot displays a data science workspace interface. On the left, the 'Explorer' panel shows a tree view of workspace resources under 'scaler-target-project-dsml', including 'Target_Business_Case_Study' and its sub-tables: 'customers', 'geolocation', 'order_items', 'order_reviews', 'orders', 'payments', 'products', and 'sellers'. The main editor area, titled 'Untitled 5', contains a SQL query to find the minimum and maximum order purchase timestamps. Below the editor, the 'Query results' section shows a table with the query's output.

JOB INFORMATION			RESULTS	JSON	EXECUTION
Row	First_Order	Last_Order			
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC			

3. Count the number of Cities and States in our dataset.

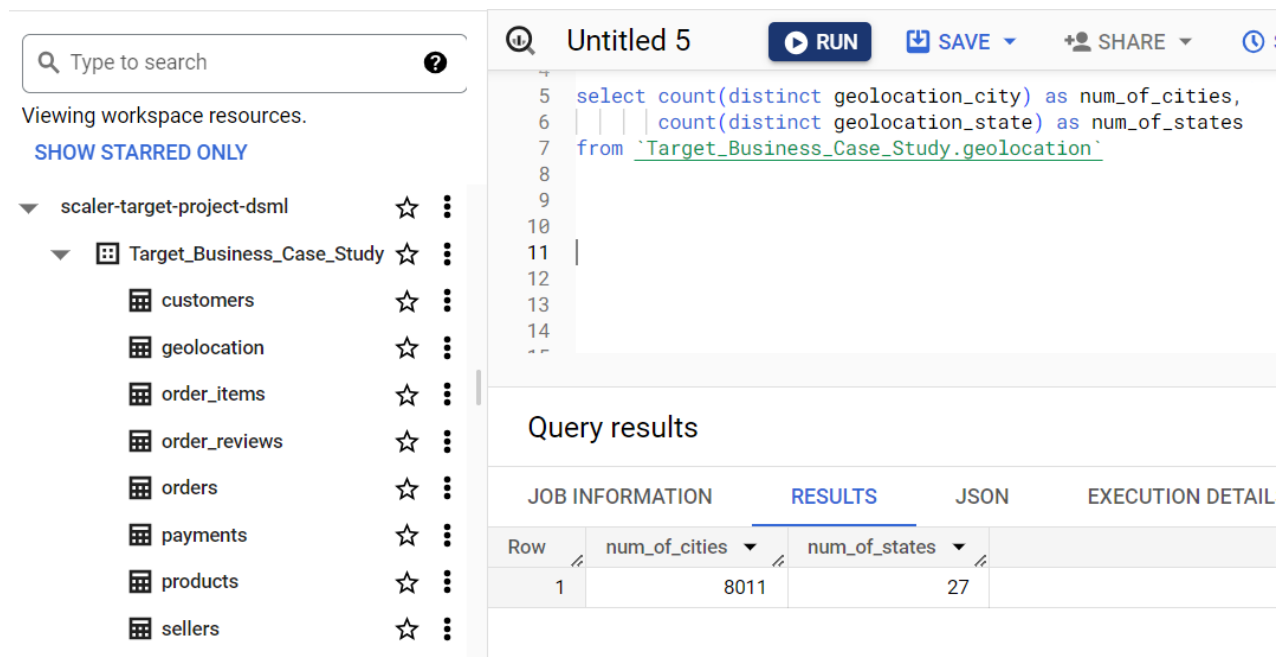
Answer:

As the question clearly mentions to count the number of Cities and States, there are 8011 cities and 27 states in the given dataset.

We can perform a count distinct on geolocation_city and geolocation_state columns from the geolocation table to extract the number of cities and states.

The Query is as follows:

```
select count(distinct geolocation_city) as num_of_cities,  
       count(distinct geolocation_state) as num_of_states  
from `Target_Business_Case_Study.geolocation`
```



The screenshot shows a SQL query editor interface. On the left is a sidebar with a search bar and a list of workspace resources under 'scaler-target-project-dsml'. The main area displays a SQL query titled 'Untitled 5' with buttons for 'RUN', 'SAVE', and 'SHARE'. Below the query editor, the 'Query results' section is visible, showing a table with the results of the query.

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAIL
Row	num_of_cities	num_of_states		
1	8011	27		

Actionable Insights & Recommendations

Recommendation:

To expand our business and attract more customers, I recommend focusing our efforts on encouraging sellers from the states of AL (Alagoas), AP (Amapá), RR (Roraima), and TO (Tocantins) to join our platform. Currently, sellers from these four states are not registered on our platform, representing a missed opportunity for growth and market presence.

Actionable Insights:

1. **Targeted Advertising Campaign:** Develop a targeted advertising campaign specifically designed to reach sellers in AL, AP, RR, and TO. This campaign should highlight the benefits of joining our platform, such as increased visibility, access to a broader customer base, and streamlined selling processes. Utilize online channels, social media platforms, and local advertising platforms to reach potential sellers in these states.
2. **Localized Marketing Materials:** Create localized marketing materials in the native languages of these states, considering cultural nuances and preferences. This approach will resonate better with potential sellers, showcasing our commitment to understanding and catering to their specific needs.
3. **Partnership with Local Organizations:** Collaborate with local trade organizations, chambers of commerce, and entrepreneurial networks in these states. Establish partnerships to promote our platform and the advantages it offers to sellers.

By focusing on these recommendations and implementing a targeted approach, we can effectively encourage sellers from AL, AP, RR, and TO to join our platform. This expansion will not only increase our seller base but also attract more customers from these states, resulting in overall business growth and increased revenue potential.

Query used to reach these Actions and insights

```
select count(distinct(seller_state)) as num_of_states
from `Target_Business_Case_Study.sellers`
```

The screenshot shows a data query interface. On the left is an 'Explorer' panel with a search bar and a tree view of workspace resources. The tree view shows a folder 'scaler-target-project-dsml' containing a folder 'Target_Business_Case_Study' which includes tables like 'customers', 'geolocation', 'order_items', 'order_reviews', 'orders', 'payments', 'products', and 'sellers'. The main panel displays a SQL query in a text editor titled 'Untitled 5'. The query is:

```
select count(distinct(seller_state)) as num_of_states
from `Target_Business_Case_Study.sellers`
```

. Below the query editor, the 'Query results' section is visible, showing a table with one row and one column 'num_of_states' with the value 23.

Row	num_of_states
1	23

This query gives us the total number of states where the sellers are registered with us

```
select distinct g.geolocation_state
from `Target_Business_Case_Study.geolocation` as g left join
`Target_Business_Case_Study.sellers` as s
on g.geolocation_state = s.seller_state
where s.seller_state is null
```

The screenshot shows a data query interface. On the left is an 'Explorer' panel with a search bar and a tree view of workspace resources. The tree view shows a folder 'scaler-target-project-dsml' containing a folder 'Target_Business_Case_Study' which includes tables like 'customers', 'geolocation', 'order_items', 'order_reviews', 'orders', 'payments', 'products', and 'sellers'. The main panel displays a SQL query in a text editor titled 'Untitled 10'. The query is:

```
select distinct g.geolocation_state
from `Target_Business_Case_Study.geolocation` as g left join `Target_Business_Case_Study.sellers` as s
on g.geolocation_state = s.seller_state
where s.seller_state is null
```

. Below the query editor, the 'Query results' section is visible, showing a table with four rows and one column 'geolocation_state' with values AL, AP, RR, and TO.

Row	geolocation_state
1	AL
2	AP
3	RR
4	TO

With the help of this query, I was able to find out the states and their names(State Code) where sellers were not registered on our platform.

2.In-depth Exploration:

1. Is there a growing trend in the no. of orders placed over the past years?

Answer:

Query to extract date and time

```
select *,
       extract(date from order_purchase_timestamp) as date,
       extract(time from order_purchase_timestamp) as time
from `Target_Business_Case_Study.orders`
```

The screenshot shows a data exploration tool interface. On the left is an 'Explorer' sidebar with a search bar and a tree view of workspace resources. The main area contains a query editor for 'question 1' with a SQL query to extract date and time from order_purchase_timestamp. Below the editor is a 'Query results' section with tabs for JOB INFORMATION, RESULTS, JSON, EXECUTION DETAILS, and EXECUTION GRAPH. The 'RESULTS' tab is active, displaying a table with 10 rows of order data. A red error message is visible at the top right of the query editor.

Syntax error: Expected end of input but got keyword SELECT at [5:1]

Row	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date	order_delivered_customer_date	order_estimated_delivery_date	date	time
1	2017-11-25 11:10:33 UTC	null	null	null	2017-12-12 00:00:00 UTC	2017-11-25	11:10:33
2	2017-12-05 01:07:58 UTC	null	null	null	2018-01-08 00:00:00 UTC	2017-12-05	01:07:58
3	2017-12-05 01:07:52 UTC	null	null	null	2018-01-11 00:00:00 UTC	2017-12-05	01:07:52
4	2018-02-09 17:21:04 UTC	null	null	null	2018-03-07 00:00:00 UTC	2018-02-09	17:21:04
5	2017-11-06 13:12:34 UTC	null	null	null	2017-12-01 00:00:00 UTC	2017-11-06	13:12:34
6	2017-04-20 12:45:34 UTC	2017-04-22 09:10:13 UTC	2017-04-24 11:31:17 UTC	null	2017-05-18 00:00:00 UTC	2017-04-20	12:45:34
7	2017-07-13 11:03:05 UTC	2017-07-13 11:10:22 UTC	2017-07-18 18:17:30 UTC	null	2017-08-14 00:00:00 UTC	2017-07-13	11:03:05
8	2017-07-11 13:36:30 UTC	2017-07-11 13:45:15 UTC	2017-07-13 17:55:46 UTC	null	2017-08-14 00:00:00 UTC	2017-07-11	13:36:30
9	2017-07-29 18:05:07 UTC	2017-07-29 18:15:17 UTC	2017-07-31 16:41:59 UTC	null	2017-08-14 00:00:00 UTC	2017-07-29	18:05:07
10	2017-07-13 10:02:47 UTC	2017-07-14 02:25:54 UTC	2017-07-20 20:02:58 UTC	null	2017-08-14 00:00:00 UTC	2017-07-13	10:02:47

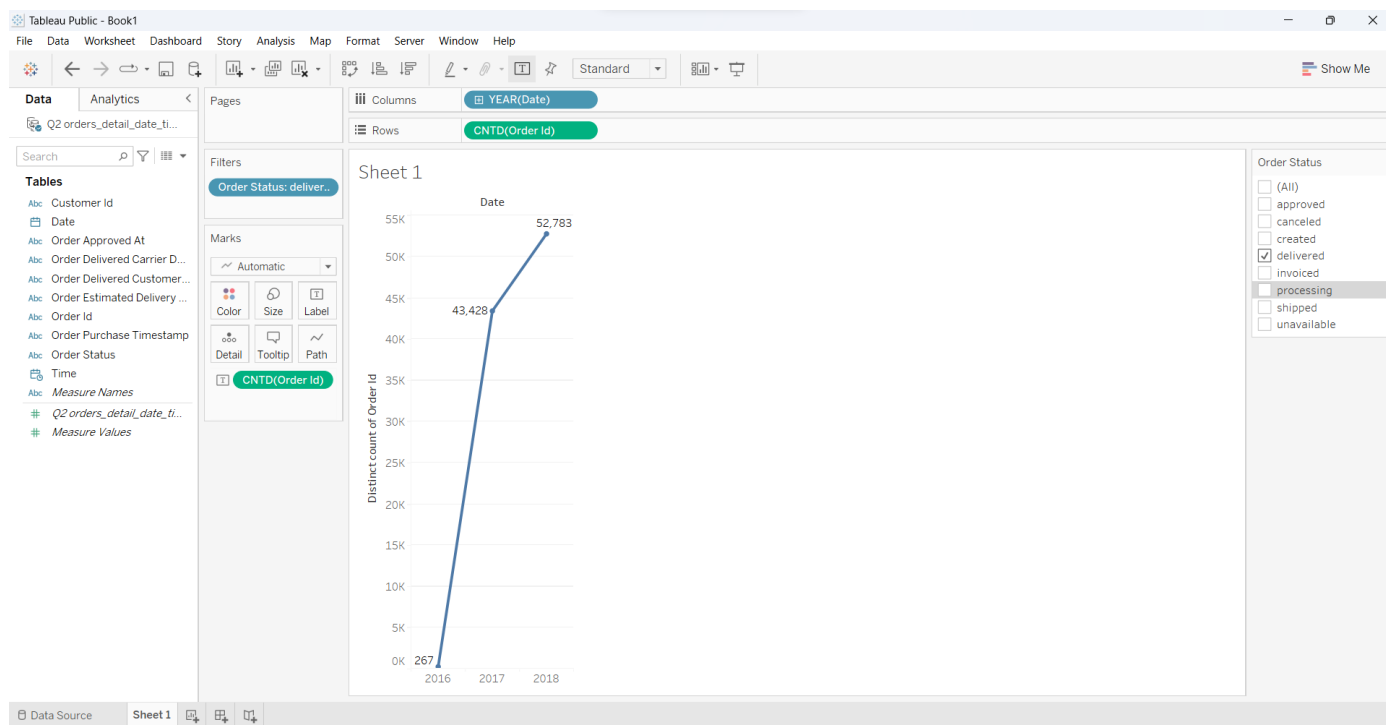
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Significant Growth in Order Placement and Delivery: 2016-2018

Considering the orders placed and delivered, there is a massive jump from the number of orders placed in 2016 to the number of orders placed in 2017.

43,428 orders were placed and delivered in 2017 compared to only 265 orders in 2016.

In the year 2018, 52,783 orders were made again seeing a significant jump from the previous year.



2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

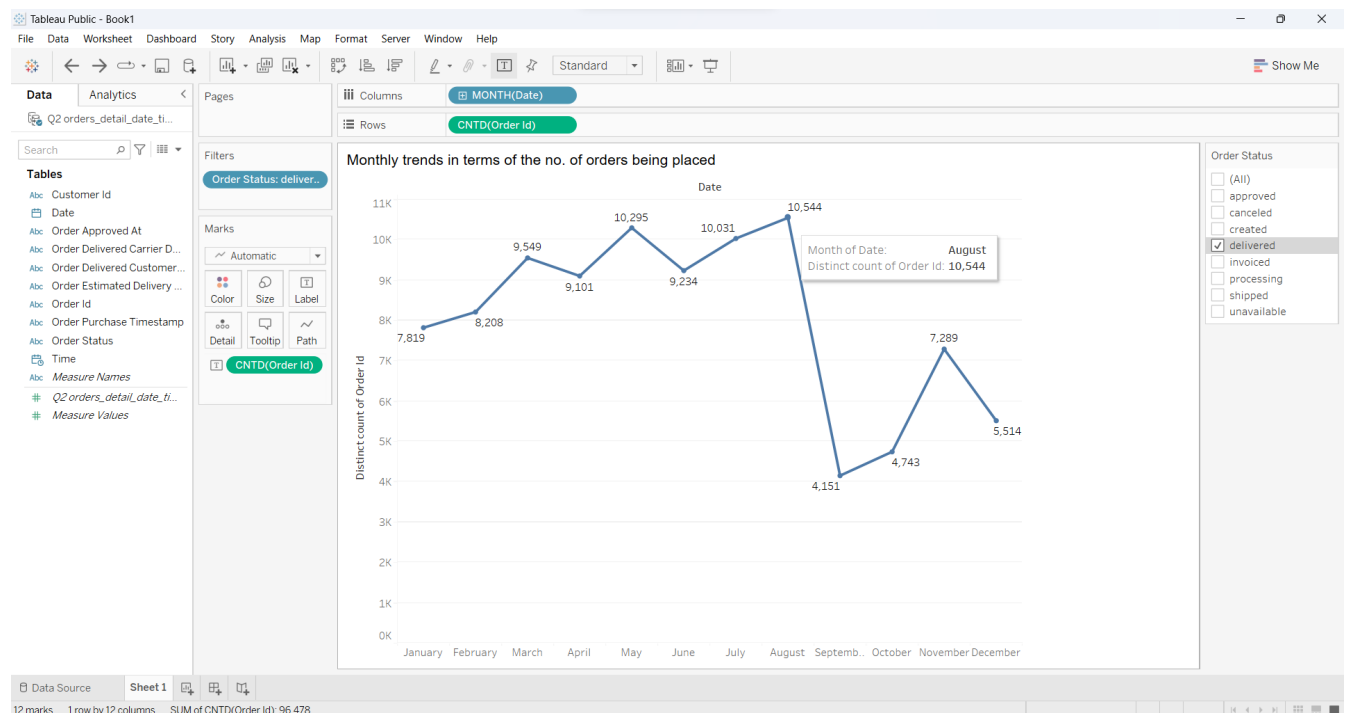
Answer:

Capturing Lucrative Celebrations: May and August Order Surge

In Brazil, the month of May records the maximum number of orders with a staggering 10,295, while August follows closely behind with 10,544 orders.

This surge in demand can be attributed to two significant occasions celebrated by Brazilians. Mother's Day, which falls on the second Sunday of May, is widely observed and cherished, making it one of the most lucrative shopping dates in the country. Similarly, in August, Brazilians celebrate Father's Day on the second Sunday of the month, honoring their dads and creating another notable shopping season. These two occasions bring about a significant boost in consumer activity and contribute to the remarkable order numbers seen in May and August.

From August to September, there is a noticeable decline in the number of orders. However, a subsequent increase in orders can be observed in the month of November, although this peak is lower in magnitude compared to the ones witnessed in May and June.



3. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon, or Night)

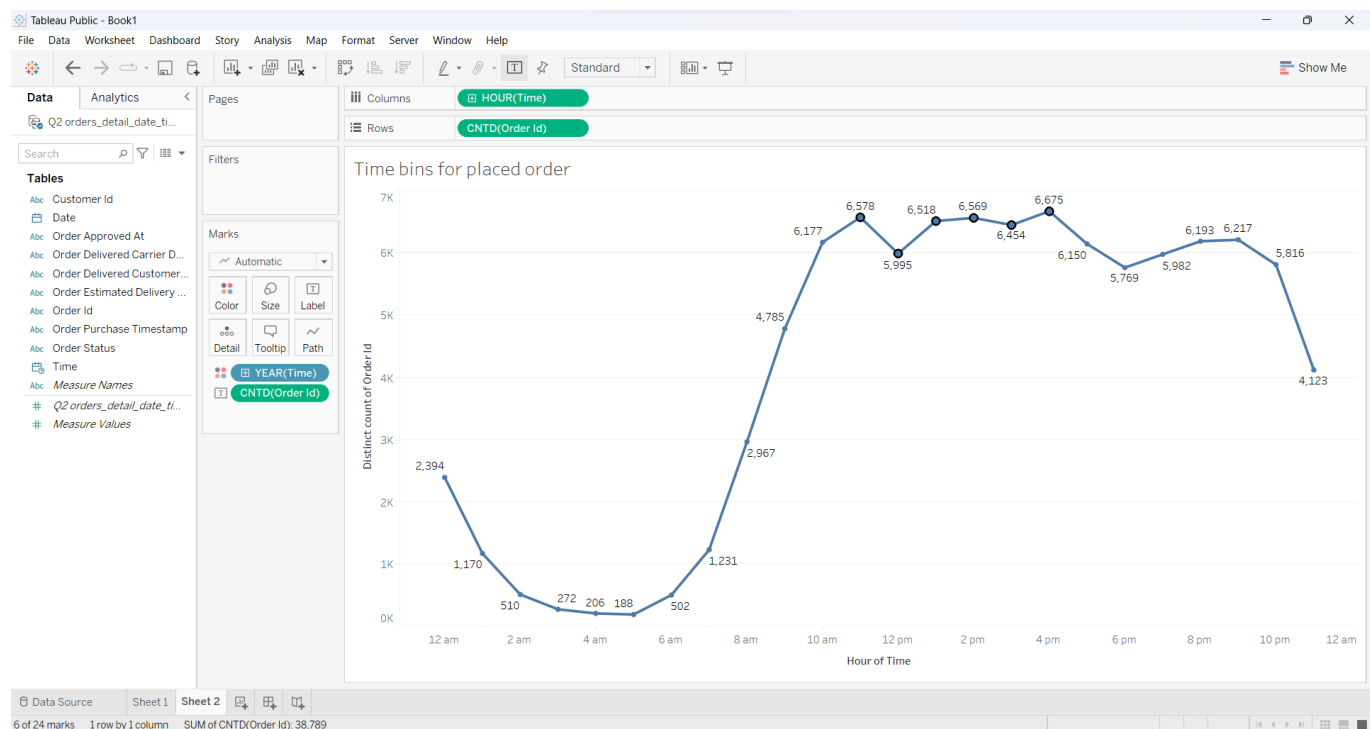
- **0-6 hrs: Dawn**
- **7-12 hrs: Mornings**
- **13-18 hrs: Afternoon**
- **19-23 hrs: Night**

Answer:

Brazilian customers mostly place their orders during the afternoon between 13 - 18 hrs.

In Brazil, the majority of customer orders are placed during the afternoon hours, specifically between 13:00 and 18:00. The morning period experiences a gradual rise in order placement, while the afternoon sees a peak in orders, maintaining a relatively steady volume until around 16:00.

This pattern highlights the preference of Brazilian customers for making purchases during the afternoon hours, indicating a potential opportunity for businesses to target and optimize their marketing and operational strategies during this time frame.



Actionable Insights & Recommendations

1. Capitalize on the growth trend: The significant increase in orders placed and delivered from 2016 to 2018 indicates a growing market and customer base. It is essential for the business to leverage this momentum and continue expanding its operations to meet the rising demand.
2. Focus on May and August: The months of May and August witness a substantial surge in orders due to the celebrations of Mother's Day and Father's Day, respectively. The business should develop targeted marketing campaigns and promotions to capitalize on these occasions and attract more customers during these periods.
3. Address the decline in orders from August to September: The noticeable decline in orders between August and September suggests a potential drop in customer engagement or interest. The business should investigate the reasons behind this decline and implement strategies to retain customers and maintain their interest during this period. Offering special promotions, and discounts, or introducing new products can help stimulate demand during this time.
4. Maximize the afternoon peak hours: The data reveals that Brazilian customers prefer to place orders during the afternoon hours, particularly between 13:00 and 18:00. The business should optimize its operational efficiency and allocate sufficient resources during this time frame to ensure smooth order processing, timely deliveries, and excellent customer service.
5. Explore opportunities in the morning hours: While the afternoon remains the peak period, there is a gradual increase in orders during the morning. The business should consider strategies to capture this growing morning demand, such as offering morning-specific promotions or introducing exclusive products or services during that time.
6. Monitor and leverage November peak: Although the November peak in orders is lower compared to May and August, it still represents an opportunity for increased sales. The business should closely monitor customer behavior during this period and develop targeted marketing initiatives, such as seasonal promotions or limited-time offers, to drive sales and boost revenue.
7. Customer engagement and retention: To sustain business growth, it is crucial to prioritize customer engagement and retention. Implementing loyalty programs, personalized offers, and excellent customer service can help foster long-term relationships with customers, leading to repeat purchases and positive word-of-mouth referrals.

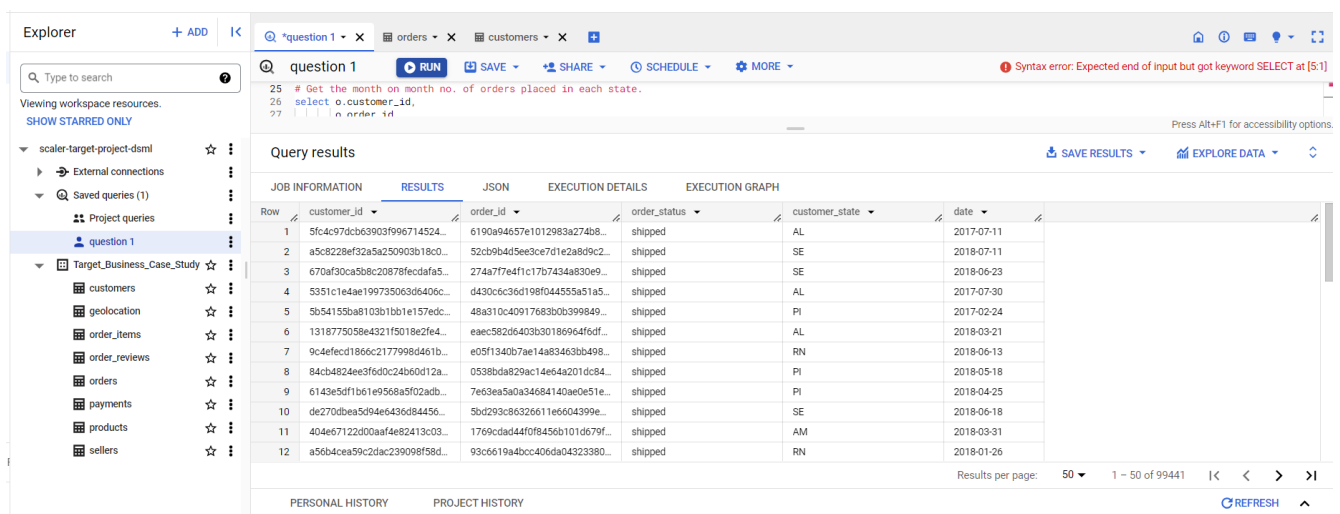
3. Evolution of E-commerce orders in the Brazil region

1. Get the month-on-month no. of orders placed in each state.

Answer

Query to get the date and State.

```
select o.customer_id,  
       o.order_id,  
       o.order_status,  
       c.customer_state,  
       extract(date from o.order_purchase_timestamp) as date  
from `Target_Business_Case_Study.orders` as o left join  
`Target_Business_Case_Study.customers` as c  
on o.customer_id = c.customer_id
```



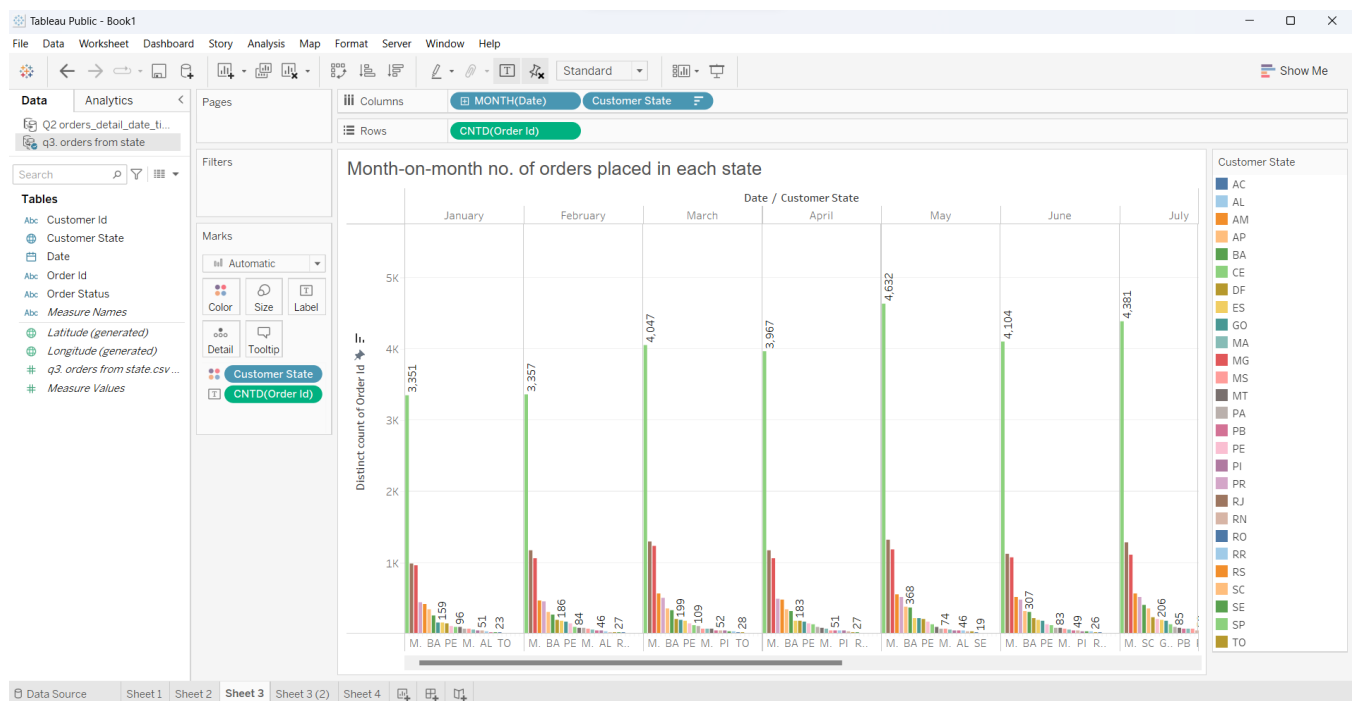
Query results

Row	customer_id	order_id	order_status	customer_state	date
1	5fc4c97dcb63903f996714524...	6190a94657e1012983a274b8...	shipped	AL	2017-07-11
2	a5c8228ef32a5a250903b18c0...	52cb9b4d5ee3ce7d1e2a8d9c2...	shipped	SE	2018-07-11
3	670af30ca5b8c20878fecda5...	274a77e4f1c17b7434a830e9...	shipped	SE	2018-06-23
4	5351c1e4ae199735063d6405c...	d430c6c36d198f044555a51a5...	shipped	AL	2017-07-30
5	5b54155ba8103b1bb1e157edc...	48a310c40917683b0b399849...	shipped	PI	2017-02-24
6	1318775058e4321f5018e2fe4...	eaec582d6403b30186964f6df...	shipped	AL	2018-03-21
7	9c4efec1866c2177998d461b...	e05f1340b7ae14a83463bb498...	shipped	RN	2018-06-13
8	84cb4824e3fed0c24b60d12a...	0538bd829ac14e64a201dc84...	shipped	PI	2018-05-18
9	6143e5df1b61e9568a5f02adb...	7e53ea5a0a34684140ae0e51e...	shipped	PI	2018-04-25
10	de270d8ea5094e6436d84456...	5bd293c86326611e6604399e...	shipped	SE	2018-06-18
11	404e67122d00aaf4e62413c03...	1769cdad44f0f8456b101d679f...	shipped	AM	2018-03-31
12	a56b4cea59c2dac239098f58d...	93c6619a4bcc406da04323380...	shipped	RN	2018-01-26

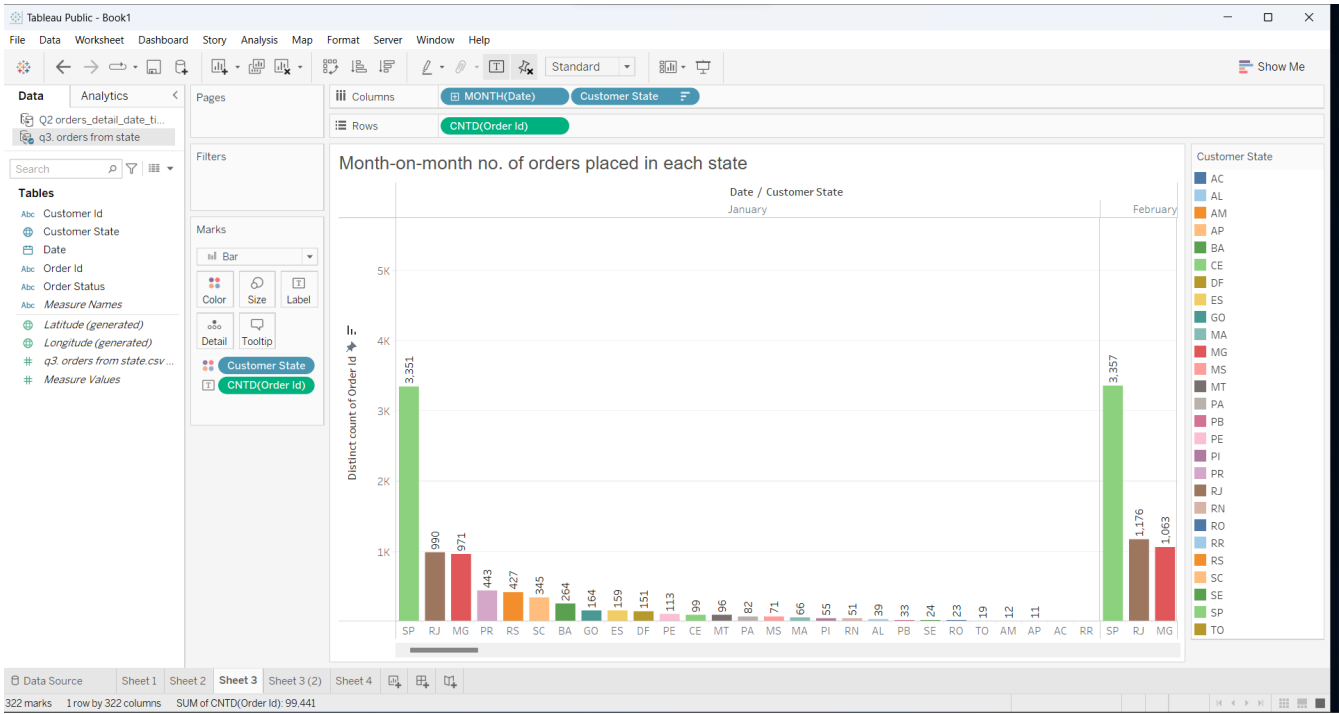
The analysis of the month-on-month number of orders placed reveals significant variations among states. Sao Paulo (SP) consistently leads with the highest number of orders placed each month, indicating a substantial volume compared to other states. Following Sao Paulo, Rio de Janeiro (RJ) consistently ranks as the state with the second-highest number of orders placed per month.

On the other hand, the states of Acre (AC), Roraima (RR), and Amapá (AP) consistently exhibit minimal to negligible order placement activity, suggesting low demand in these regions.

This data underscores the importance of Sao Paulo and Rio de Janeiro as key markets for the business. These states demonstrate a robust customer base and present lucrative opportunities for revenue generation. Conversely, the states of Acre, Roraima, and Amapá may require further examination to understand the reasons behind the limited order placement and explore strategies to potentially stimulate demand in these regions.



The number of orders in January month for each State



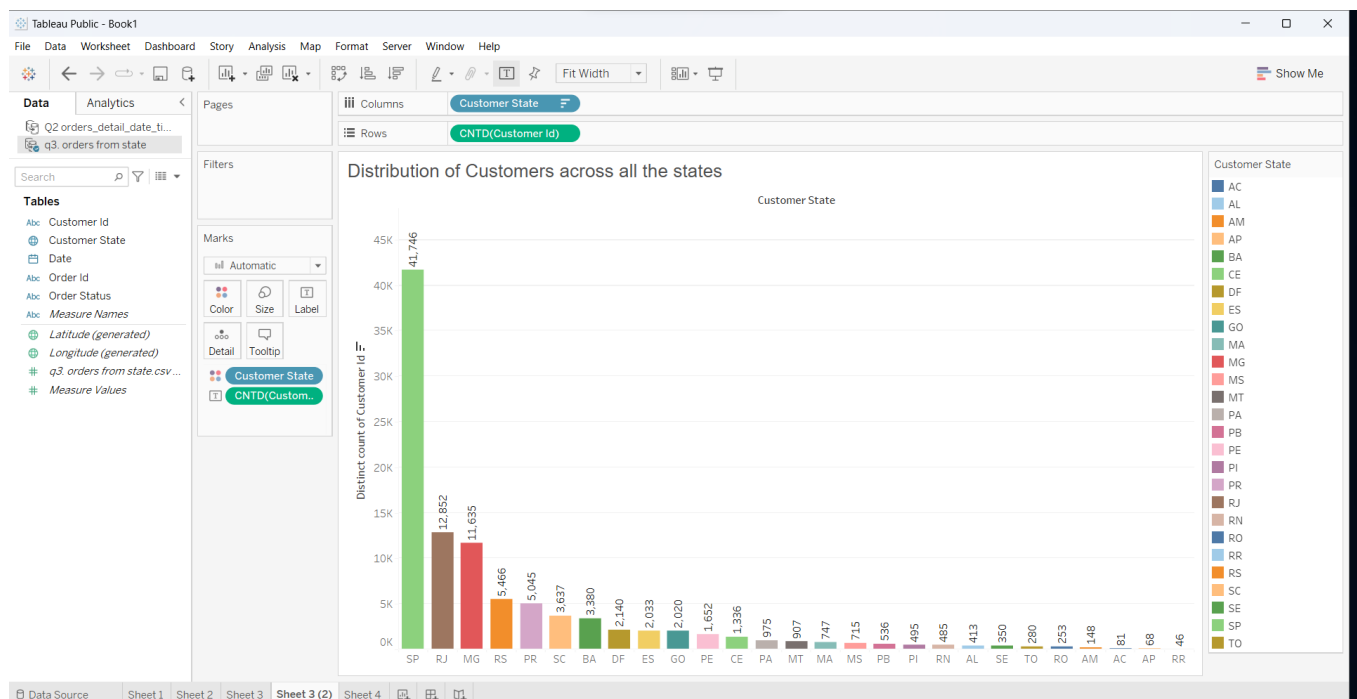
2. How are the customers distributed across all the states?

The distribution of customers across all states in Brazil reveals notable variations. Sao Paulo (SP) state has the highest number of customers, with a total of 41,746. Following Sao Paulo, Rio de Janeiro (RJ) state holds the second-largest customer base, with 12,852 customers.

In addition to Sao Paulo and Rio de Janeiro, the state of Minas Gerais (MG) also exhibits a significant customer base, contributing to the overall distribution of customers.

Conversely, the states of Acre (AC), Amapa (AP), and Roraima (RR) have the lowest customer bases in Brazil, indicating comparatively lower customer populations in these regions.

This information is valuable for businesses as it provides insights into the geographic distribution of their customer base. Understanding the concentration of customers across different states helps organizations focus their marketing efforts, tailor product offerings, and allocate resources accordingly.



Actionable Insights & Recommendations

1. Focus on Sao Paulo and Rio de Janeiro: Sao Paulo and Rio de Janeiro consistently show high order placement numbers and have significant customer bases. These states present lucrative opportunities for business growth. Allocate marketing resources, optimize logistics, and tailor product offerings to cater to the preferences and demands of customers in these regions.
2. Explore opportunities in Minas Gerais: Minas Gerais exhibits a notable customer base and can be a potential market for business expansion. Conduct market research to understand the specific needs and preferences of customers in this state, and develop targeted marketing campaigns to attract and engage them.
3. Investigate low-demand states: Acre, Roraima, and Amapa demonstrate low order placement activity and have relatively smaller customer bases. Conduct a detailed analysis to understand the underlying factors contributing to this low demand. Explore potential strategies to stimulate demand in these regions, such as targeted promotions, local partnerships, or customized offerings based on the specific needs of customers in these states.
4. Implement state-specific marketing strategies: Since customer preferences and behaviors may vary across states, consider implementing state-specific marketing strategies. This could involve tailoring messaging, product offerings, and promotions to align with the cultural, social, and economic characteristics of each state. For example, conducting market research to identify specific preferences or trends in Sao Paulo and designing targeted campaigns based on that research.
5. Collaborate with local influencers or partners: In low-demand states or states with untapped potential, consider collaborating with local influencers or establishing partnerships with businesses in those regions. This can help raise brand awareness, generate interest, and build credibility among the local customer base.

4. Impact on the Economy: Analyze the money movement by e-commerce by looking at order prices, freight, and others.

1. Get the % increase in the cost of orders from the year 2017 to 2018 (include months between Jan to Aug only).

You can use the "payment_value" column in the payments table to get the cost of orders.

Answer

The Query is as follows:

```
SELECT
    ROUND(((SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018 THEN
op.payment_value END)
    - SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017 THEN
op.payment_value END))
    / SUM(CASE WHEN EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2017 THEN
op.payment_value END)) * 100,2) AS percentage_increase

FROM `Target_Business_Case_Study.orders` as o JOIN
`Target_Business_Case_Study.payments` as op
ON o.order_id = op.order_id

WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
```

The screenshot displays a data analytics tool interface. On the left, the 'Explorer' panel shows a workspace with resources like 'scaler-target-project-dsml', 'External connections', 'Saved queries (1)', 'Project queries', 'question 1', and 'Target_Business_Case_Study' (which includes 'customers', 'geolocation', and 'order_items'). The main area shows a SQL query in 'Untitled 11' with a 'RUN' button. The query is the same as the one provided in the text. Below the query editor, the 'Query results' section is visible, showing a table with one row and one column, 'percentage_increase', with a value of 136.98. The interface also includes a top navigation bar with tabs for 'orders', 'products', 'order_items', and 'Untitled11', and a bottom status bar indicating 'Query completed'.

Row	percentage_increase
1	136.98

the % increase in the cost of orders from the year 2017 to 2018 (include months between Jan to Aug only) is **136.98%**

*question 1 ✕ orders ✕ *Untitled20 ✕ customers ✕ order_items ✕ *Untitled23 ✕ +

Untitled 23

RUN SAVE SHARE SCHEDULE MORE This query

```

1 SELECT DISTINCT c.customer_state,
2     ROUND(SUM(oi.price),2) AS total_order_price,|
3     ROUND(AVG(oi.price),2) AS average_order_price
4 FROM `Target_Business_Case_Study.customers` c JOIN `Target_Business_Case_Study.orders` o ON c.customer_id = o.customer_id
5 JOIN `Target_Business_Case_Study.order_items` oi ON o.order_id = oi.order_id

```

Query results

SAVE RESULTS

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	customer_state	total_order_price	average_order_price		
1	SP	5202955.05	109.65		
2	RJ	1824092.67	125.12		
3	MG	1585308.03	120.75		
4	RS	750304.02	120.34		
5	PR	683083.76	119.0		
6	SC	520553.34	124.65		
7	BA	511349.99	134.6		
8	DF	302603.94	125.77		
9	GO	294591.95	126.27		
10	ES	275037.31	121.91		

Results per page: 50 1 - 2

3. Calculate the Total & Average value of order freight for each state.

Answer:

The Query is as follows:

```
SELECT c.customer_state,
       ROUND(SUM(oi.freight_value),2) AS total_order_freight,
       ROUND(AVG(oi.freight_value),2) AS average_order_freight
FROM `Target_Business_Case_Study.customers` c JOIN
`Target_Business_Case_Study.orders` o ON c.customer_id = o.customer_id
JOIN `Target_Business_Case_Study.order_items` oi ON o.order_id = oi.order_id
GROUP BY c.customer_state
ORDER BY average_order_freight DESC
```

*question 1	orders	*Untitled20	customers	order_items	*Untitled23	+
Untitled 23						
RUN SAVE SHARE SCHEDULE MORE						
<pre>1 SELECT c.customer_state, 2 ROUND(SUM(oi.freight_value),2) AS total_order_freight, 3 ROUND(AVG(oi.freight_value),2) AS average_order_freight 4 FROM `Target_Business_Case_Study.customers` c JOIN `Target_Business_Case_Study.orders` o ON c.customer_id = o.customer_id 5 JOIN `Target_Business_Case_Study.order_items` oi ON o.order_id = oi.order_id</pre>						
Query results						
SAVE RESULTS						
JOB INFORMATION RESULTS JSON EXECUTION DETAILS EXECUTION GRAPH						
Row	customer_state	total_order_freight	average_order_freight			
1	RR	2235.19	42.98			
2	PB	25719.73	42.72			
3	RO	11417.38	41.07			
4	AC	3686.75	40.07			
5	PI	21218.2	39.15			
6	MA	31523.77	38.26			
7	TO	11732.68	37.25			
8	SE	14111.47	36.65			
9	AL	15914.59	35.84			
10	PA	38699.3	35.83			

Actionable Insights & Recommendations

1. Focus on states with high average order freight: States such as RR (Roraima), PB(Paraiba) and RO (Rondonia) have relatively higher average order freight values. This suggests potential areas for optimizing shipping and logistics operations to reduce costs and improve efficiency. Explore strategies like negotiating better freight rates with shipping providers or optimizing packaging to minimize dimensional weight.
2. The states with low average order freight: States like MG (Minas Gerais), PR (Parana), and SP (Sao Paulo) have comparatively lower average order freight values. Investigate the reasons behind this, as it could indicate opportunities to improve cost-effective shipping methods or leverage partnerships with local carriers for more affordable and efficient delivery options.
3. Explore local fulfillment centers: For states with high average order freight values, consider setting up local fulfillment centers or partnering with third-party logistics providers (3PLs) in strategic locations. This can help reduce shipping distances, optimize delivery times, and potentially lower overall freight costs.
4. Negotiate shipping contracts: Evaluate existing shipping contracts and negotiate with shipping providers to secure more favorable rates based on order volumes, frequency, and destinations. Consolidating shipments or partnering with other businesses in the same region to achieve higher shipping volumes can provide leverage in negotiations.

5. Analysis based on sales, freight, and delivery time.

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- $\text{time_to_deliver} = \text{order_delivered_customer_date} - \text{order_purchase_timestamp}$
- $\text{diff_estimated_delivery} = \text{order_estimated_delivery_date} - \text{order_delivered_customer_date}$

Answer:

The Query is as follows:

```
SELECT order_id,
       order_purchase_timestamp,
       order_delivered_customer_date,
       order_estimated_delivery_date,
       DATE_DIFF(order_delivered_customer_date , order_purchase_timestamp ,
day) as delivery_time,
       DATE_DIFF(order_estimated_delivery_date ,
order_delivered_customer_date , day) as estimated_delivery_diff
FROM `Target_Business_Case_Study.orders`
```

*question 1

orders

question 1

RUN

SAVE

SHARE

SCHEDULE

MORE

```
88 SELECT order_id,
89         order_purchase_timestamp,
90         order_delivered_customer_date,
91         order_estimated_delivery_date,
92         DATE_DIFF(order_delivered_customer_date , order_purchase_timestamp , day) as delivery_time,
```

Press Alt+F1 for accessibility of

Query results

SAVE RESULTSEXPLORE DATA

JOB INFORMATION

RESULTS

JSON

EXECUTION DETAILS

EXECUTION GRAPH

Row	order_id	order_purchase_timestamp	order_delivered_customer_date	order_estimated_delivery_date	delivery_time	estimated_delivery_diff
1	1950d777989f6a877539f5379...	2018-02-19 19:48:52 UTC	2018-03-21 22:03:51 UTC	2018-03-09 00:00:00 UTC	30	-12
2	2c45c33d2f9cb8ff8b1c86cc28...	2016-10-09 15:39:56 UTC	2016-11-09 14:53:50 UTC	2016-12-08 00:00:00 UTC	30	28
3	65d1e226dfaeb8cdc42f66542...	2016-10-03 21:01:41 UTC	2016-11-08 10:58:34 UTC	2016-11-25 00:00:00 UTC	35	16
4	635c894d068ac37ee03dc54e...	2017-04-15 15:37:38 UTC	2017-05-16 14:49:55 UTC	2017-05-18 00:00:00 UTC	30	1
5	3b97562c3aee8bdedcb5c2e45...	2017-04-14 22:21:54 UTC	2017-05-17 10:52:15 UTC	2017-05-18 00:00:00 UTC	32	0
6	68f47f50f04c4cb6774570cfd...	2017-04-16 14:56:13 UTC	2017-05-16 09:07:47 UTC	2017-05-18 00:00:00 UTC	29	1
7	276e9ec344d3bf029ff83a161c...	2017-04-08 21:20:24 UTC	2017-05-22 14:11:31 UTC	2017-05-18 00:00:00 UTC	43	-4
8	54e1a3c2b97fb0809da548a59...	2017-04-11 19:49:45 UTC	2017-05-22 16:18:42 UTC	2017-05-18 00:00:00 UTC	40	-4
9	fd04fa4105ee8045f6a0139ca5...	2017-04-12 12:17:08 UTC	2017-05-19 13:44:52 UTC	2017-05-18 00:00:00 UTC	37	-1
10	302bb8109d097a9fc6e9cefc5...	2017-04-19 22:52:59 UTC	2017-05-23 14:19:48 UTC	2017-05-18 00:00:00 UTC	33	-5

Results per page: 501 – 50 of 99441|<>

2. Find out the top 5 states with the highest & lowest average freight value.

ANSWER:

Top 5 states with the highest average freight Value

```
SELECT x.customer_state,
       x.average_order_freight,
       DENSE_RANK() OVER(ORDER BY x.average_order_freight DESC ) as RANK
FROM(SELECT c.customer_state,
          ROUND(AVG(oi.freight_value),2) AS average_order_freight,
          FROM `Target_Business_Case_Study.customers` c JOIN
`Target_Business_Case_Study.orders` o ON c.customer_id = o.customer_id
        JOIN `Target_Business_Case_Study.order_items` oi ON o.order_id = oi.order_id
        GROUP BY c.customer_state) as x
ORDER BY RANK ASC
LIMIT 5
```

question 1	RUN	SAVE	SHARE	SCHEDULE	MORE
102	SELECT	x.customer_state,			
103		x.average_order_freight,			
104		DENSE_RANK() OVER(ORDER BY x.average_order_freight DESC) as RANK			
105	FROM(SELECT	c.customer_state,			
106		ROUND(AVG(oi.freight_value),2) AS average_order_freight,			
107		FROM `Target_Business_Case_Study.customers` c JOIN `Target_Business_Case_Study.orders` o ON c.customer_id = o.customer_id			
108		JOIN `Target_Business_Case_Study.order_items` oi ON o.order_id = oi.order_id			
109		GROUP BY c.customer_state) as x			
110	ORDER BY	RANK ASC			
111	LIMIT	5			
112					
113					
114					

Query results

SAVE RESULTS

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	customer_state	average_order_freigh	RANK		
1	RR	42.98	1		
2	PB	42.72	2		
3	RO	41.07	3		
4	AC	40.07	4		
5	PI	39.15	5		

Top 5 states with the Lowest average freight Value

```
SELECT x.customer_state,
       x.average_order_freight,
       DENSE_RANK() OVER(ORDER BY x.average_order_freight ASC ) as RANK
FROM(SELECT c.customer_state,
          ROUND(AVG(oi.freight_value),2) AS average_order_freight,
          FROM `Target_Business_Case_Study.customers` c JOIN
`Target_Business_Case_Study.orders` o ON c.customer_id = o.customer_id
        JOIN `Target_Business_Case_Study.order_items` oi ON o.order_id = oi.order_id
        GROUP BY c.customer_state) as x
ORDER BY RANK ASC
LIMIT 5
```

The screenshot displays a data analytics platform interface. On the left is an 'Explorer' sidebar with a search bar and a list of workspace resources including 'scaler-target-project-dsml', 'External connections', 'Saved queries (1)', 'Project queries', and 'question 1'. Below these are database tables: 'Target_Business_Case_Study', 'customers', 'geolocation', 'order_items', 'order_reviews', 'orders', 'payments', 'products', and 'sellers'. The main panel shows a SQL query for 'question 1' with a 'RUN' button and options to 'SAVE', 'SHARE', 'SCHEDULE', and 'MORE'. The query is a SQL statement to find the top 5 states with the lowest average freight value. Below the query editor, the 'Query results' section is visible, showing a table with columns 'customer_state', 'average_order_freight', and 'RANK'. The results are as follows:

Row	customer_state	average_order_freight	RANK
1	SP	15.15	1
2	PR	20.53	2
3	MG	20.63	3
4	RJ	20.96	4
5	DF	21.04	5

3. Find out the top 5 states with the highest & lowest average delivery time.

ANSWER:

Top 5 states with the highest average delivery time are

```
SELECT x.*,
       DENSE_RANK() OVER(ORDER BY x.average_delivery_time DESC) as Rank
FROM(SELECT c.customer_state,
          ROUND(AVG(DATE_DIFF(order_delivered_customer_date,
order_purchase_timestamp, day)),2) AS average_delivery_time
      FROM `Target_Business_Case_Study.orders` as o join
`Target_Business_Case_Study.customers` as c
      ON o.customer_id = c.customer_id
      GROUP BY c.customer_state) as x
ORDER BY Rank ASC
LIMIT 5
```

The screenshot displays a SQL IDE interface. On the left is an 'Explorer' pane showing a project structure with tables like 'customers', 'geolocation', 'order_items', 'order_reviews', 'orders', 'payments', and 'products'. The main editor shows a SQL query titled 'Untitled 20' that calculates the average delivery time for each customer state and ranks them. Below the query editor, the 'Query results' section shows a table with 5 rows of data.

Row	customer_state	average_delivery_time	Rank
1	RR	28.98	1
2	AP	26.73	2
3	AM	25.99	3
4	AL	24.04	4
5	PA	23.32	5

Top 5 states with the Lowest average delivery time are

```
SELECT x.*,
       DENSE_RANK() OVER(ORDER BY x.average_delivery_time ASC) as Rank
FROM(SELECT c.customer_state,
          ROUND(AVG(DATE_DIFF(order_delivered_customer_date,
order_purchase_timestamp, day)),2) AS average_delivery_time
       FROM `Target_Business_Case_Study.orders` as o join
`Target_Business_Case_Study.customers` as c
       ON o.customer_id = c.customer_id
       GROUP BY c.customer_state) as x
ORDER BY Rank ASC
LIMIT 5
```

question 1 RUN SAVE SHARE SCHEDULE MORE

```
140 -- top 5 states with the Lowest average delivery time
141 SELECT x.*,
142        DENSE_RANK() OVER(ORDER BY x.average_delivery_time ASC) as Rank
143 FROM(SELECT c.customer_state,
144          ROUND(AVG(DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, day)),2) AS average_delivery_time
145        FROM `Target_Business_Case_Study.orders` as o join `Target_Business_Case_Study.customers` as c
146        ON o.customer_id = c.customer_id
147        GROUP BY c.customer_state) as x
148 ORDER BY Rank ASC
149 LIMIT 5
150
```

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Query results SAVE RESULTS EXPLORE DATA

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	customer_state	average_delivery_time	Rank		
1	SP	8.3	1		
2	PR	11.53	2		
3	MG	11.54	3		
4	DF	12.51	4		
5	SC	14.48	5		

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

ANSWER

```
SELECT x.*,
       DENSE_RANK() OVER(ORDER BY x.delivery_diff ASC) as Rank
FROM(SELECT c.customer_state,
          ROUND(AVG(DATE_DIFF(order_estimated_delivery_date ,
order_delivered_customer_date , day)),2) AS delivery_diff
       FROM `Target_Business_Case_Study.orders` as o join
`Target_Business_Case_Study.customers` as c
       ON o.customer_id = c.customer_id
       GROUP BY c.customer_state) as x
ORDER BY Rank ASC
LIMIT 5
```

question 1

RUNSAVESHARESCHEDULEMORE

154

155 SELECT x.*,

156 DENSE_RANK() OVER(ORDER BY x.delivery_diff ASC) as Rank

157 FROM(SELECT c.customer_state,

158 ROUND(AVG(DATE_DIFF(order_estimated_delivery_date , order_delivered_customer_date , day)),2) AS delivery_diff

159 FROM `Target\_Business\_Case\_Study.orders` as o join `Target\_Business\_Case\_Study.customers` as c

160 ON o.customer_id = c.customer_id

161 GROUP BY c.customer_state) as x

162 ORDER BY Rank ASC

163 LIMIT 5

164 |

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Query results

SAVE RESULTSEXPLORE DATA

JOB INFORMATIONRESULTSJSONEXECUTION DETAILSEXECUTION GRAPH

Row	customer_state	delivery_diff	Rank
1	AL	7.95	1
2	MA	8.77	2
3	SE	9.17	3
4	ES	9.62	4
5	BA	9.93	5

PERSONAL HISTORYPROJECT HISTORY

REF

Actionable Insights & Recommendations

1. Focus on states with high average freight value: States like RR, PB, RO, AC, and PI have higher average freight values. Explore opportunities to optimize shipping costs in these states by negotiating better rates with shipping providers, exploring alternative shipping methods, or optimizing packaging to reduce dimensional weight.
2. Evaluate logistics operations: States with high average delivery times, such as RR, AP, AM, AL, and PA, may indicate potential challenges in logistics operations. Identify bottlenecks in the delivery process, assess carrier performance, and streamline internal processes to improve delivery efficiency. Consider partnering with reliable and efficient carriers or implementing technology solutions to track and manage shipments more effectively.
3. Optimize delivery estimation accuracy: States where the order delivery is faster compared to the estimated date of delivery, such as AL, MA, SE, ES, and BA, indicate efficient delivery operations. Leverage this advantage by improving delivery estimation accuracy for all states. Implement systems and processes to provide customers with realistic and reliable estimated delivery dates, which can help manage expectations and enhance customer satisfaction.
4. Expand warehouse or fulfillment center network: To improve delivery efficiency, consider establishing additional warehouses or fulfillment centers strategically located in states with higher average delivery times. This allows for quicker order processing, reduced transit times, and improved customer satisfaction.
5. Enhance customer communication: Maintain transparent and proactive communication with customers throughout the delivery process. Provide timely updates on order status, tracking information, and any potential delays. Effective communication can help manage customer expectations and reduce customer inquiries, leading to improved satisfaction.

6. Optimize shipping methods: Assess different shipping methods and carriers to identify the most cost-effective and reliable options. Consider offering multiple shipping options to customers, including expedited shipping for those who require faster delivery. Implement systems to track and compare carrier performance, ensuring timely and reliable deliveries.

6. Analysis based on the payments:

1. Find the month-on-month no. of orders placed using different payment types.

ANSWER:

The query is as follows:

```
SELECT distinct(p.payment_type),
       FORMAT_TIMESTAMP('%B', o.order_purchase_timestamp) AS Month,
       COUNT(p.order_id) over(partition by p.payment_type order by EXTRACT(month
FROM o.order_purchase_timestamp)RANGE BETWEEN 0 PRECEDING AND 0
       FOLLOWING) as Num_of_orders
FROM `Target_Business_Case_Study.payments` as p LEFT JOIN
`Target_Business_Case_Study.orders` o
ON p.order_id = o.order_id
```

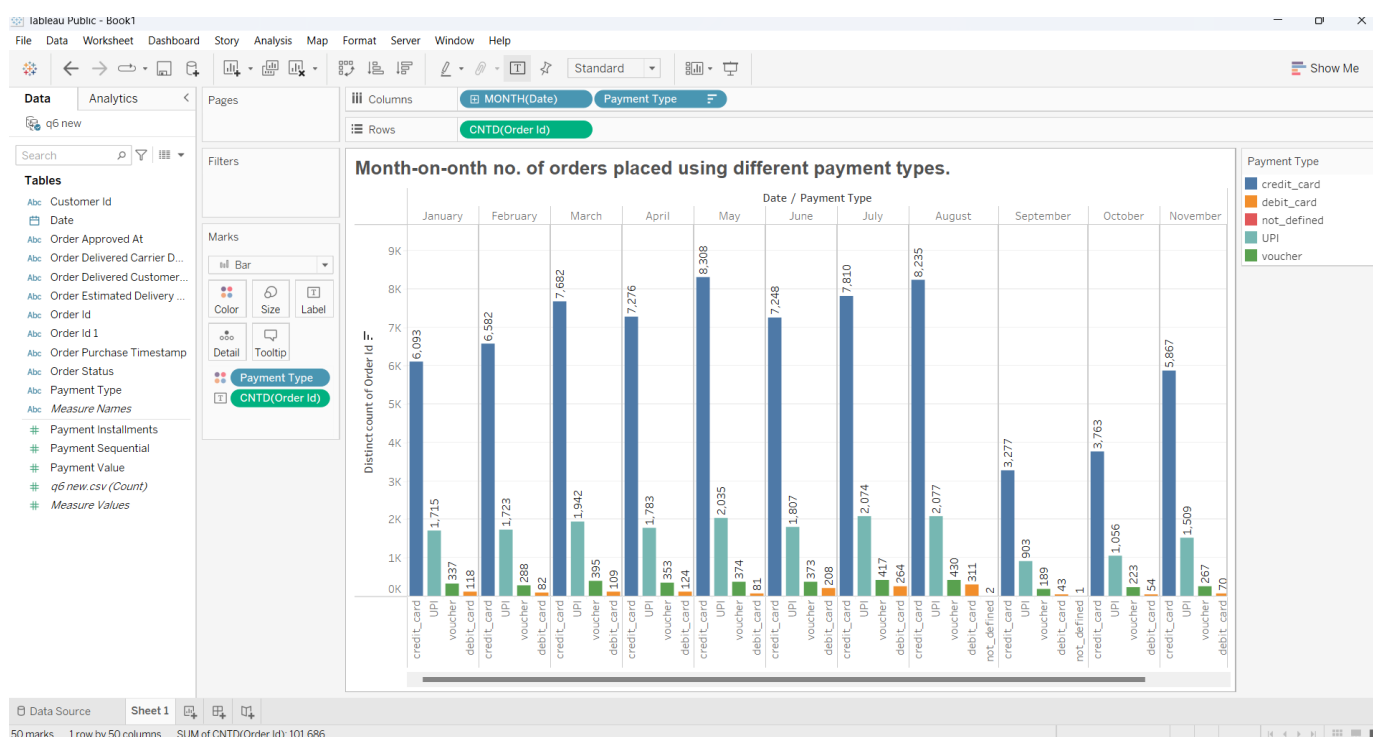
question 1	RUN	SAVE	SHARE	SCHEDULE	MORE
167	SELECT distinct(p.payment_type),				
168	FORMAT_TIMESTAMP('%B', o.order_purchase_timestamp) AS Month,				
169	COUNT(p.order_id) over(partition by p.payment_type order by EXTRACT(month FROM o.order_purchase_timestamp)RANGE BETWEEN 0 PRECEDING AND 0 FOLLOWING) as Num_of_orders				
	FROM `Target_Business_Case_Study.payments` as p LEFT JOIN				
	`Target_Business_Case_Study.orders` o				
	ON p.order_id = o.order_id				

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	payment_type	Month	Num_of_orders		
1	UPI	January	1715		
2	UPI	February	1723		
3	UPI	March	1942		
4	UPI	April	1783		
5	UPI	May	2035		
6	UPI	June	1807		
7	UPI	July	2074		
8	UPI	August	2077		
9	UPI	September	903		
10	UPI	October	1056		
11	UPI	November	1509		
12	UPI	December	1160		

```
question 1
RUN SAVE SHARE SCHEDULE MORE
167 SELECT distinct(p.payment_type),
168         FORMAT_TIMESTAMP('%B', o.order_purchase_timestamp) AS Month,
169         COUNT(p.order_id) over(partition by p.payment_type order by EXTRACT(month FROM
```

Query results

JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS	EXECUTION GRAPH
Row	payment_type	Month	Num_of_orders		
12	UPI	December	1160		
13	credit_card	January	6103		
14	credit_card	February	6609		
15	credit_card	March	7707		
16	credit_card	April	7301		
17	credit_card	May	8350		
18	credit_card	June	7276		
19	credit_card	July	7841		
20	credit_card	August	8269		
21	credit_card	September	3286		
22	credit_card	October	3778		
23	credit_card	November	5897		



From this data, we can infer that credit card payments account for the highest number of transactions. UPI (Unified Payments Interface) holds the second position, while the least number of payments are made through other means. Notably, there is a significant surge in payments across all payment categories observed between the months of March and August.

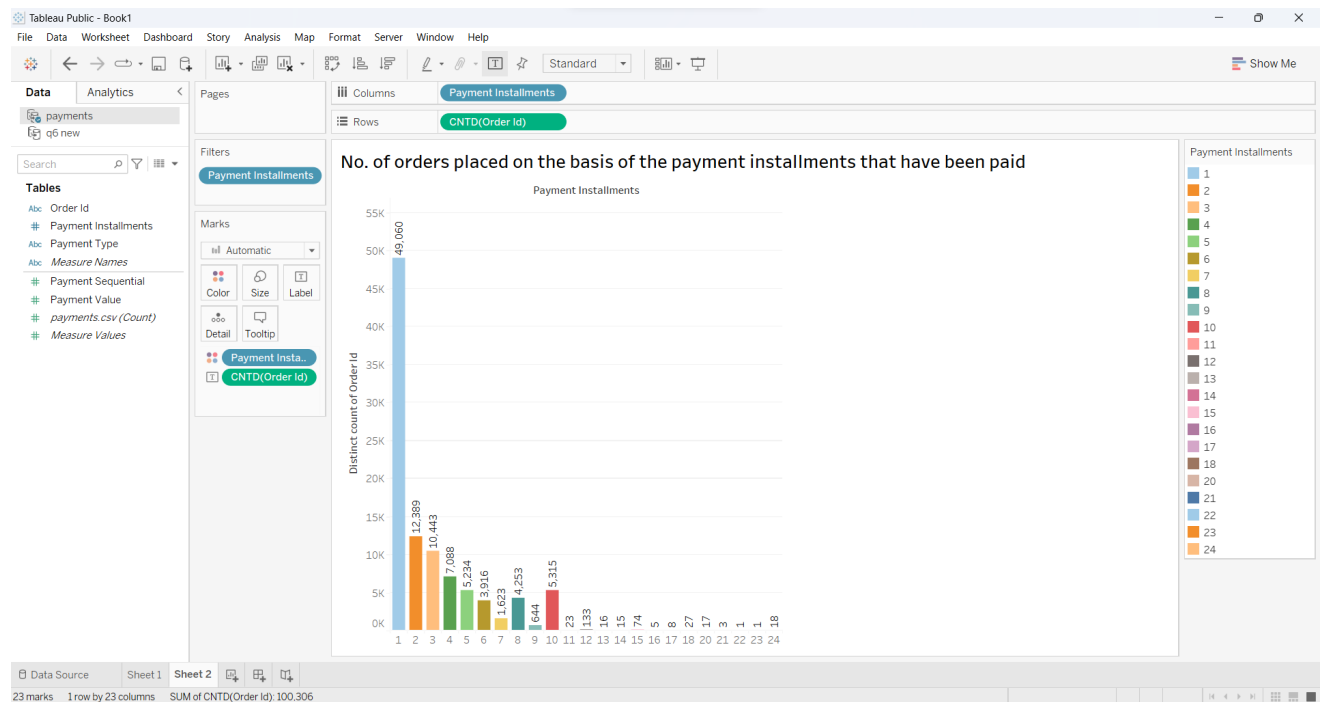
2. Find the no. of orders placed on the basis of the payment installments that have been paid.

ANSWER:

The Query is as follows:

```
SELECT payment_installments,  
       COUNT(DISTINCT order_id) as Num_orders  
FROM `Target_Business_Case_Study.payments`  
WHERE payment_installments >= 1  
GROUP BY payment_installments
```

*question 1 X payments X orders X +				
question 1 RUN SAVE SHARE (				
175 SELECT payment_installments, 176 COUNT(DISTINCT order_id) as Num_orders 177 FROM `Target_Business_Case_Study.payments` 178 WHERE payment_installments >= 1 179 GROUP BY payment_installments				
Query results				
JOB INFORMATION		RESULTS	JSON	EXECUTION DETAILS
Row	payment_installment	Num_orders		
1	1	49060		
2	2	12389		
3	3	10443		
4	4	7088		
5	5	5234		
6	6	3916		
7	7	1623		
8	8	4253		
9	9	644		
10	10	5315		



Actionable Insights & Recommendations:

1. Capitalize on credit card dominance: Since credit card payments account for the highest number of transactions, it is essential to prioritize and optimize this payment method. Ensure seamless credit card processing, secure payment gateways, and a user-friendly checkout experience to encourage more customers to choose credit cards for payments.
2. Promote UPI as an alternative: While UPI holds the second position in terms of the number of payments, there is potential for further growth. Educate customers about the benefits and convenience of UPI payments, such as faster transactions, no need for physical cards, and interoperability across different banks. Consider offering incentives, discounts, or exclusive promotions for customers who choose to pay via UPI.
3. Streamline other payment methods: Although the number of payments made through other means is relatively low, it is still

important to ensure a smooth and efficient payment experience for customers who prefer these methods. Evaluate the user experience and optimize the checkout process for these payment options, addressing any potential pain points or barriers that may discourage customers from using them.

4. Leverage the surge in payments from March to August: The significant increase in payments across all categories during this period presents an opportunity to capitalize on heightened customer activity. Consider running targeted marketing campaigns, offering exclusive deals or discounts, and optimizing inventory management to meet the increased demand. Additionally, analyze customer behavior during this period to identify patterns and preferences that can inform future business strategies.